

Package ‘blstinypplot’

May 8, 2026

Title Plots for Seasonal Adjustment Analysts using tinypplot

Version 1.1

Description

Generates several types of time series plots useful for seasonal adjustment analysis using ggplot2. These routines rely heavily on the seasonal package to extract series and components from the 'seasonal adjustments generated by the US Census Bureau's X-13ARIMA-SEATS software, and can be generated from a single seas object or a list of seas objects. Types of plots include line plots, ratio plots, forecast plots, seasonal factor plots, seasonal adjustment component plots. Users can add grid lines, remove background shading, and shade recession regions in selected plots.

License MIT + file LICENSE

Encoding UTF-8

LazyData TRUE

Roxygen list(markdown = TRUE)

RoxygenNote 8.0.0

Imports lubridate,
plotrix,
RColorBrewer,
Rdpack,
seasonal (>= 1.10.0),
stringr (>= 1.5.0),
tinypplot,
tis

Suggests zoo

Depends R (>= 4.0.0)

RdMacros Rdpack

Contents

add_outlier_lines	2
cnv_color_codes	3
color_blind_palette	4
convert_date_to_tis	5
display_color	6
draw_recession	7

employment_data_mts	8
employment_list	8
flag_peak	9
gen_tick_positions	10
get_auto_outlier_string	11
get_month_index	12
get_recession_dates	12
get_reg_string	13
get_udg_index	14
plot_ratio	15
plot_series	17
plot_sf	20
plot_table	22
proc_outlier	25
reset_par	26
set_day_and_month	27
shoes2008	27
visual_sig_peaks	28
xt_data_list	29

Index 30

add_outlier_lines	<i>Plot table from X-13ARIMA-SEATS seasonal adjustment run.</i>
-------------------	---

Description

Generate plot of user-specified table from the output of a seas object generated by the seasonal package.

Usage

```
add_outlier_lines(
  seas_obj = NULL,
  otl_col = NULL,
  add_sub_title = TRUE,
  sub_title_cex = 0.7
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
otl_col	Character array of length 6; color used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is colors from the RColorBrewer palette "Paired" (Neuwirth 2022).
add_sub_title	Logical scalar; indicates if plots will include subtitles denoting what series are plotted. Default includes a subheader.
sub_title_cex	Numeric scalar; scaling for main title of plot. Default is 0.7.

Details

Version 1.3, 2026-04-08

Value

Outlier lines are added to an existing tinyplot plot.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
shoes_seas <-
  seasonal::seas(shoes2008, x11 = "",
    transform.function = "log",
    regression.aictest = c("td", "easter"),
    outlier.types = "all",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 60)
shoes_sa <- seasonal::final(shoes_seas)
plot_series(shoes_sa,
  y_axis_label = "Shoes",
  do_grid = TRUE,
  draw_recess = TRUE,
  this_col = "black",
  start_plot = c(1958,1),
  this_point_type = 1,
  main_title = "X-11 Adjustment for Shoe Store Sales",
  main_title_line = 1.5,
  main_title_cex = 1.0,
  this_reset = TRUE)
add_outlier_lines(shoes_seas)
```

cnv_color_codes

Generates color names from hexadecimal input

Description

Generates vector of closest color names from hexadecimal color input. This uses the `color.id` function of the `plotrix` package (Lemon 2006) to do the conversion.

Usage

```
cnv_color_codes(color_vec = NULL)
```

Arguments

color_vec vector of color codes This is a required entry.

Details

Version 1.1, 2026-04-01

Value

vector of color names closest to hexadecimal color input.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Lemon J (2006). "Plotrix: a package in the red light district of R." *R-News*, **6**(4), 8-12.

Examples

```
Moonrise_Codes <-
  c("#F3DF6C", "#CEAB07", "#D5D5D3", "#24281A", "#798E87",
    "#C27D38", "#CCC591", "#29211F", "#85D4E3", "#F4B5BD",
    "#9C964A", "#CDC08C", "#FAD77B")
Moonrise_All <- cnv_color_codes(Moonrise_Codes)
```

color_blind_palette *Color-blind friendly color palette*

Description

Color palettes that can be used that can be distinguished by color-blind people (either from Neuwirth (2022) or Chang (2025)).

Usage

```
color_blind_palette(with_grey = TRUE)
```

Arguments

with_grey Logical scalar; whether color blind pallate contains 'grey', otherwise the palette contains black. Default is TRUE.

Details

Version 1.1, 2026-04-01

Value

vector of hexadecimal color codes that form a color palette that can be distinguished by color-blind people.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Chang W (2025). “Colors (ggplot2) — cookbook-r.com.” [http://www.cookbook-r.com/Graphs/Colors_\(ggplot2\)/](http://www.cookbook-r.com/Graphs/Colors_(ggplot2)/).

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Examples

```
this_color_blind <- color_blind_palette(FALSE)
```

convert_date_to_tis	<i>Convert ts dates to tis format</i>
---------------------	---------------------------------------

Description

Convert dates used for monthly (or quarterly) ts series to tis (Hallman 2021) formats.

Usage

```
convert_date_to_tis(
  this_date,
  this_freq = 12,
  is_start = TRUE,
  return_tis = FALSE
)
```

Arguments

this_date	numeric scalar or vector; ts date to be converted
this_freq	numeric scalar; frequency of ts time series. Default is 12 (monthly).
is_start	logical scalar; is date assumed to be the beginning of the month? Default is TRUE; if FALSE, date is assumed to be at the end of the month.
return_tis	logical scalar; If true, return as tis object; otherwise return as integer. Default is FALSE.

Details

Version 1.0, 2026-04-01

Value

A tis index value that is the equivalent of the ts date.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hallman J (2021). *tis: Time Indexes and Time Indexed Series*. R package version 1.39, <https://CRAN.R-project.org/package=tis>.

Examples

```
this_month <- as.numeric(substr(Sys.Date(),1,4)) +
  (as.numeric(substr(Sys.Date(),6,7)) - 1) / 12
end_this_month_tis <-
  convert_date_to_tis(this_month, this_freq = 12,
                      is_start = FALSE, return_tis = TRUE)
start_this_month_tis <-
  convert_date_to_tis(this_month, this_freq = 12,
                      is_start = TRUE, return_tis = TRUE)
```

display_color	<i>Color name for display</i>
---------------	-------------------------------

Description

Generates color names for display on plot labels and subheaders.

Usage

```
display_color(this_color_code = NULL, strip_numbers = FALSE)
```

Arguments

this_color_code	character string of color code to be used in plot This is a required entry.
strip_numbers	logical scalar that controls if numbers at the end of the text are stripped from the color name. Default is TRUE.

Details

Version 1.1, 2026-04-07

Value

character string of color name closest to hexadecimal color input (if used) stripped of numbers if strip_numbers = TRUE.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_color_blind <- color_blind_palette()
this_color_blind_text <- array(NA, dim = 8)
for (i in 1:8) {
  this_color_blind_text[i] <- display_color(this_color_blind[i])
}
```

draw_recession	<i>Draw NBER recessions</i>
----------------	-----------------------------

Description

Draws shaded areas in plots corresponding to NBER recessions (NBER 2026).

Usage

```
draw_recession(
  this_col_recess = NULL,
  this_density = 50,
  this_border = NA,
  this_add_recess_start = NULL,
  this_sub_recess = TRUE,
  this_sub_line = 2.5,
  this_sub_cex = 0.75,
  display_color_strip = FALSE
)
```

Arguments

<code>this_col_recess</code>	Character string; color used for shading recession periods. Default is 'lightgrey'.
<code>this_density</code>	the density of shading lines, in lines per inch. The default value is 50. A zero value of density means no shading lines whereas negative values (and NA) suppress shading (and so allow color filling).
<code>this_border</code>	Integer scalar; thickness of border around region. Default is NA, meaning the border is not generated.
<code>this_add_recess_start</code>	numeric scalar; Starting date for an additional recession period at the end of the series. Default is not to add recession dates.
<code>this_sub_recess</code>	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
<code>this_sub_line</code>	Integer scalar; position of subtitle of plot. Default is 2.5.
<code>this_sub_cex</code>	Numeric scalar; scaling for subtitle of plot. Default is 0.75.
<code>display_color_strip</code>	Logical scalar; indicates if the display color will be stripped of trailing numbers. Default is FALSE.

Details

Version 1.0, 2026-04-01

Value

Shades recession dates in plots generated by `tinypplot` (McDermott et al. 2025).

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    x11='')
plot_table(air_seas, 'd12', 'AirPassengers',
  do_grid = TRUE,
  draw_recess = FALSE,
  use_ratio = FALSE,
  add_sub_title = TRUE,
  this_col = 'forestgreen')
draw_recession(this_col_recess = 'lightblue',
  this_border = 1,
  this_sub_line = 2.25,
  this_sub_cex = 0.75)
```

employment_data_mts	<i>US Unemployment Series, four main components in an mts object</i>
---------------------	--

Description

An mts object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
data(employment_data_mts)
```

Format

An mts object with 4 time series elements in four columns:

n2000013 Employed Males 16-19

n2000014 Employed Females 16-19

n2000025 Employed Males 20+

n2000026 Employed Females 20+

employment_list	<i>US Employment Series, four main components in a list object.</i>
-----------------	---

Description

A list object of the four main components of US Employment expressed as time series objects that end in December, 2022.

Usage

```
data(employment_list)
```


Format

A list object with 4 time series elements:

n2000013 Employed Males 16-19

n2000014 Employed Females 16-19

n2000025 Employed Males 20+

n2000026 Employed Females 20+

flag_peak	<i>Flag visual significant peaks in spectra</i>
-----------	---

Description

Determine positions of visual significant peaks in spectra using the criteria of Soukup and Findley (1999).

Usage

```
flag_peak(seas_obj = NULL, spec_type = NULL, spec_freq_code = NULL)
```

Arguments

seas_obj seas object generated from a call of the seas of the seasonal package on a single time series. This is a required argument.

spec_type Character string; type of spectrum. Possible values are 'ori', 'irr', 'rsd', 'sa'.

spec_freq_code Character string; type of frequency being tested. Possible values are 's' or 't'.

Details

Version 1.0, 2026-04-07

Value

If visually significant peaks found, a numeric vector of the position of the peak frequencies. If no peaks found, 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Soukup RJ, Findley DF (1999). "On the Spectrum Diagnostics Used by X-12-ARIMA to Indicate the Presence of Trading Day Effects after Modeling or Adjustment." *Proceeding of the American Statistical Association, Business and Economic Statistics Section*, 144–149. <https://www.census.gov/content/dam/Census/library/working-papers/1999/adrm/rr9903s.pdf>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)', x11='')
this_flagged_peak_seas <- flag_peak(air_seas,'ori','s')
this_flagged_peak_td <- flag_peak(air_seas,'ori','t')
```

gen_tick_positions	<i>Generate tick mark positions</i>
--------------------	-------------------------------------

Description

Generate tick mark position for x or y axis for base R plots

Usage

```
gen_tick_positions(
  this_series = NULL,
  tick_interval = 2,
  is_date = FALSE,
  time_interval = "years",
  this_base = 1
)
```

Arguments

this_series	numeric or date vector, the values used for the x-axis. This is a required entry.
tick_interval	integer scalar; the value separating the tick marks. Default is 2.
is_date	logical scalar; indicates if this_series is a vector of Date objects. Default is FALSE, this_series is numeric.
time_interval	character scalar; the time unites used to construct the x-axis if this_series is a Date object vector. Default is "years".
this_base	numeric scalar; number used to integer divide this_series for numeric vectors. Default is 1.

Details

Version 1.2, 2026-04-06

Value

Generates values for major tick marks for the X and Y axis of plots.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```

air_time <- time(AirPassengers)
air_Date <-
  as.Date(paste(as.numeric(floor(time(AirPassengers))),
               as.numeric(cycle(AirPassengers)),
               "1", sep = "-"))

air_x_axis_time <-
  gen_tick_positions(air_time, tick_interval = 3)
air_x_axis_Date <-
  gen_tick_positions(air_Date, tick_interval = 3, is_date = TRUE)
air_y_axis <-
  gen_tick_positions(AirPassengers, tick_interval = 200,
                    this_base = 100)

```

get_auto_outlier_string

Get automatic outlier names

Description

Get the names of outliers identified in the seas object for a single series.

Usage

```
get_auto_outlier_string(seas_obj = NULL, return_string = TRUE)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package. This is a required entry.
return_string	Logical scalar; if TRUE, results returned as a character string (default). Else, results returned as a character array.

Details

Version 1.0, 2026-04-08

Value

Character string (or array if return_string is TRUE) containing a summary of the outliers identified in the regARIMA model. If no regressors or automatic outliers in the model, the routine will return a blank character (or blank vector).

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```

air_seas <-
  seasonal::seas(AirPassengers, arima.model = "(0 1 1)(0 1 1)", x11="")
this_auto_outlier <- get_auto_outlier_string(air_seas)

```

get_month_index	<i>Generate index of month abbreviation</i>
-----------------	---

Description

Process string of month abbreviation to return a numeric index .

Usage

```
get_month_index(this_month_string)
```

Arguments

this_month_string
Character string; 3 character abbreviation of month

Details

Version 1.0, 2026-04-07

Value

Index of month - 1 for 'Jan', 2 for 'Feb', etc.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this0t1 <- 'A02015.Jan'
thisCode <- 'A0'
thisPerChar <- substr(this0t1,nchar(thisCode)+6,nchar(this0t1))
thisPerIndex <- get_month_index(thisPerChar)
```

get_recession_dates	<i>Get NBER recession dates</i>
---------------------	---------------------------------

Description

Add shading for US NBER recession dates (NBER 2026). Recession dates are taken from the nberDates function of the tis package (Hallman 2021).

Usage

```
get_recession_dates(
  start_recess = NULL,
  end_recess = NULL,
  add_recess_start = NULL,
  this_freq = 12
)
```

Arguments

start_recess numeric scalar; Starting date for plot. Default is first recession starting date.

end_recess numeric scalar; Ending date for plot. Default is last recession ending date.

add_recess_start numeric scalar; Starting date for an additional recession period at the end of the series. Default is not to add recession dates.

this_freq numeric scalar; frequency of ts time series. Default is 12 (monthly).

Details

Version 1.0, 2026-04-01

Value

Starting and ending dates for NBER recessions within a span of data

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Hallman J (2021). *tis: Time Indexes and Time Indexed Series*. R package version 1.39, <https://CRAN.R-project.org/package=tis>.

NBER (2026). “Business Cycle Dating.” <https://www.nber.org/research/business-cycle-dating>.

Examples

```
plot_limits <- c(1949, 1961)
thisRec <-
  get_recession_dates(start_recess = plot_limits[1],
                      end_recess = plot_limits[2])
```

get_reg_string	<i>Get names of regressors</i>
----------------	--------------------------------

Description

Generate string of names for the regressors used in the model fit for a given series.

Usage

```
get_reg_string(seas_obj = NULL, xreg_names = NULL, exclude_user = FALSE)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
xreg_names	Character vector with names of user defined regressors used in model. Default is NULL, no user defined regressors. Number of names in this vector should match number of user-defined regressors; if not, a warning message will be produced.
exclude_user	Logical scalar; determines if user defined regressors are included in the string of regression names output by the function. Default is FALSE

Details

Version 1.1, 2026-04-07

Value

Character string containing a summary of the regressors in the regARIMA model. If no regressors in the model, the routine will return a blank character.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    transform.function = "log",
    forecast.maxlead=36,
    check.print = c( "pacf", "pacfplot" ),
    x11 = "", slidingspans = "")
air_reg <- get_reg_string(air_seas)
```

get_udg_index

Index for entry in UDG list

Description

Return index for entry in UDG list returned by the udg() function of the seasonal package.

Usage

```
get_udg_index(udg_list = NULL, this_key = NULL)
```

Arguments

udg_list	List object generated by udg function of the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_key	Keyword found in udg files generated by X-13ARIMA-SEATS. This is a required entry.

Details

Version 3.1, 2026-03-05

Value

An integer denoting which element in the udg output matches the key provided by the user. If there is no match, the function returns the number 0.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
ukgas_seas <-
  seasonal::seas(UKgas,
    series.period = 4,
    transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)",
    forecast.maxlead = 20,
    check.print = c( "pacf", "pacfplot" ),
    x11 = "")
ukgas_udg <- seasonal::udg(ukgas_seas)
ukgas_udg_index_a <-
  get_udg_index(ukgas_udg, this_key='a')
```

plot_ratio

Ratio plot

Description

Generates a high-definition plot around a reference line other than zero.

Usage

```
plot_ratio(
  ratio_series = NULL,
  ratio_range = range(ratio_series),
  main_title = "Ratio Plot",
  main_title_line = 2,
  main_title_cex = NULL,
  ratio_mean = 1,
  ratio_color = "black",
  ratio_mean_color = "black",
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  x_axis_label = "Time",
  y_axis_label = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(4, 4, 4, 0.5),
  this_reset = TRUE
)
```

Arguments

ratio_series	Time series of ratios/factors for which you want to generate a high definition plot. This is a required entry.
ratio_range	Range of values you wish the plot to be plotted over. Default is range of the series.
main_title	Title for the plot. Default is character string 'Ratio Plot'.
main_title_line	Integer scalar; position of main title of plot. Default is 2.
main_title_cex	Numeric scalar; scaling for main title of plot. Default is this_plot_cex + 0.25.
ratio_mean	Assumed mean value for the ratio. Default is 1.0
ratio_color	Color used for lines in ratio plot. Default is 'black'.
ratio_mean_color	Color used for mean line in ratio plot. Default is 'black'.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the axis function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
x_axis_label	Character string; x-axis label for plot. Default is 'Time'.
y_axis_label	Character string; y-axis label for plot, if specified.
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(4, 4, 4, 0.5).
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 1.5, 2026-04-02

Value

Generates a high definition plot of rations centered on one, by default.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    transform.function= 'log',
    arima.model = '(0 1 1)(0 1 1)',
    seats.save = 's10')
air_sf <- seasonal::series(air_seas, 's10')
plot_ratio(air_sf,
  main_title = 'SEATS seasonal for Airline Passenger',
  ratio_color = 'darkblue')
```

plot_series	<i>Plot individual series.</i>
-------------	--------------------------------

Description

Generate plot of user-specified series.

Usage

```
plot_series(
  this_series = NULL,
  main_title = NULL,
  main_title_line = 2.75,
  main_title_cex = 1.25,
  x_axis_label = NULL,
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  y_limit = NULL,
  start_plot = NULL,
  do_grid = FALSE,
  draw_recess = FALSE,
  recess_start = NULL,
  recess_col = "lightgrey",
  recess_sub = TRUE,
  this_trans = TRUE,
  use_ratio = FALSE,
  this_col = "grey",
  this_plot_type = "1",
```

```

    this_line_type = 1,
    this_line_width = 1,
    this_point_type = NULL,
    this_plot_cex = 0.8,
    this_lab_cex = NULL,
    this_axis_cex = NULL,
    this_mar = c(4, 4, 4, 0.5),
    this_reset = TRUE
)

```

Arguments

<code>this_series</code>	Numeric vector; time series object to be plotted. This is a required entry.
<code>main_title</code>	Character string; main title of plot. Default is no title.
<code>main_title_line</code>	Integer scalar; position of main title of plot. Default is 2.75.
<code>main_title_cex</code>	Numeric scalar; scaling for main title of plot. Default is 1.25.
<code>x_axis_label</code>	Character string; x-axis label for plot, if specified.
<code>y_axis_label</code>	Character string; y-axis label for plot, if specified.
<code>x_axis_main_ticks</code>	Numeric vector; X-axis tick marks. Default is determined by the axis function.
<code>y_axis_main_ticks</code>	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
<code>y_limit</code>	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of <code>this_series</code> .
<code>start_plot</code>	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
<code>do_grid</code>	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
<code>draw_recess</code>	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates (NBER 2026). Default is no recession shading.
<code>recess_start</code>	numeric matrix; Rows of dates for additional recession starting and ending dates. Default is not to add recession dates.
<code>recess_col</code>	Character string; color used for shading of recession region. Default is "lightgrey".
<code>recess_sub</code>	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
<code>this_trans</code>	Logical scalar; indicates if the adjustment was done with a log transform. Default is TRUE.
<code>use_ratio</code>	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
<code>this_col</code>	Character string; color used for series. Default is 'grey'.
<code>this_plot_type</code>	Character string; type of plot produced. Typical values are "l" (lines only), "p" (points only), "b" (both points and lines), "o" (overplotted points and lines), "n" (no plotting). Default is "l".
<code>this_line_type</code>	Integer scalar; line type used for series. Default is 1.
<code>this_line_width</code>	Integer scalar; line width used for series. Default is 1.

this_point_type	Integer scalar; point type used for series. Default is no points plotted.
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(4, 4, 4, 0, 5).
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 2.0, 2026-04-08

Value

Generate plot of user-specified series. Can be first in a series of plots, with other lines or points added after calling this routine. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

NBER (2026). "Business Cycle Dating." <https://www.nber.org/research/business-cycle-dating>.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = "(0 1 1)(0 1 1)",
    outlier.types = "all",
    x11 = "",
    forecast.maxlead = 36)
air_sa <- seasonal::final(air_seas)
plot_series(air_sa,
  y_axis_label = "AirPass",
  do_grid = TRUE,
  draw_recess = TRUE,
  this_col = "black",
  start_plot = c(1958, 1),
  this_point_type = 1,
  main_title = "X-11 Seasonal Adjustment for Airline Passengers",
  main_title_line = 1.5,
  main_title_cex = 1.0,
  this_reset = TRUE)
```

plot_sf

*Seasonal factor (and the SI-ratios) plot grouped by month/quarter***Description**

Generates a special plot of the seasonal factors (and the SI-ratios) grouped by month/quarter

Usage

```
plot_sf(
  seas_obj = NULL,
  this_table = NULL,
  add_si = FALSE,
  main_title = "Seasonal Sub-Plots",
  sub_title = NULL,
  y_label = NULL,
  y_limit = NULL,
  x_label = " ",
  this_col = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_main_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(2.1, 1.5, 3.1, 0.5),
  num_facet_rows = NULL,
  do_grid = FALSE,
  this_facet_bg = "grey90",
  this_facet_cex = 0.6,
  this_tinytheme = "minimal",
  this_reset = TRUE
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddelbuettel 2018). This is a required entry.
this_table	Character string; X-13ARIMA-SEATS table name or abbreviation (such as d10 or s10). Valid table names for this argument can be found in the documentation for the series function of the seasonal package (Sax 2018). If not a valid table name, the function will print an error message and return a NULL. This entry is required.
add_si	Logical scalar; indicates if seasonal factor plots will include the SI ratios and unmodified SI ratios for X-11 seasonal adjustments. Default is not including SI ratios (add_si = FALSE).
main_title	Character string; main title of plot. Default is 'Seasonal Sub-Plots'.
sub_title	Character string; a subtitle for the plot. Default is NULL, which generates a subtitle which denotes the series that are plotted.
y_label	Character string; y-axis label for plot, if specified.

y_limit	Numeric vector of length 2; Range of values to be plotted over. Default is range of the seasonal factors.
x_label	Character string; label for x-axis of plot. Default is a blank x-axis.
this_col	Character array of length 2 or 4; color used for seasonal factors, SI-ratios, Modified SI-ratios, and seasonal means. Default is generated from the RColorBrewer palette "Dark2", with "darkgrey" added for the seasonal mean line. For a listing of R color codes, see R Coder (2025).
this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_main_cex	Numeric scalar; scaling for main plot title. Default is the value of this_plot_cex + 0.1.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(2.1, 1.5, 3.1, 0.5).
num_facet_rows	Integer scalar; number of rows in the facet plot. Default is generated from the series frequency (2 for monthly series, 1 otherwise).
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
this_facet_bg	Character scalar; Color used for the facet labels. Default is "grey90".
this_facet_cex	Integer scalar; size used for the facet labels. Default is 0.6.
this_tinytheme	Character scalar; setting for the tinythere function from the tinyplot package (McDermott et al. 2025). Default is "minimal".
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is TRUE.

Details

Version 1.3, 2026-05-08

Value

Generates a special plot of the seasonal factors (and the SI-ratios) grouped by month/quarter.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

McDermott G, Arel-Bundock V, Zeileis A (2025). *tinyplot: Lightweight Extension of the Base R Graphics System*. doi:10.32614/CRAN.package.tinyplot, R package version 0.6.0.

R Coder (2025). "Colors in R — r-charts.com." <https://r-charts.com/colors/>. [Accessed 17-02-2026].

Sax C (2018). *seasonal: R Interface to X-13ARIMA-SEATS*. R package version 1.7. <https://CRAN.R-project.org/package=seasonal>.

Sax C, Eddelbuettel D (2018). "Seasonal Adjustment by X-13ARIMA-SEATS in R." *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <- seasonal::seas(AirPassengers, arima.model = '(0 1 1)(0 1 1)',
                           x11='', x11.save = c("d8", "d9"))
plot_sf(air_seas, add_si = TRUE,
        main_title = 'Air Passengers Seasonal Sub-Plots',
        sub_title = "SF = Cyan, SI = Orange, ModSI = Purple, Mean = Gray",
        do_grid = TRUE)
```

plot_table

Plot table from X-13ARIMA-SEATS seasonal adjustment run.

Description

Generate plot of user-specified table from the output of a seas object generated by the seasonal package.

Usage

```
plot_table(
  seas_obj = NULL,
  this_table = NULL,
  main_title = NULL,
  main_title_line = 2,
  main_title_cex = 1.25,
  x_axis_label = NULL,
  y_axis_label = NULL,
  x_axis_main_ticks = NULL,
  y_axis_main_ticks = NULL,
  y_limit = NULL,
  start_plot = NULL,
  do_grid = FALSE,
  draw_recess = FALSE,
  recess_start = NULL,
  recess_col = NULL,
  recess_sub = TRUE,
  add_otl = FALSE,
  use_ratio = FALSE,
  add_sub_title = FALSE,
  sub_title_line = 1,
  sub_title_cex = NULL,
  this_line_type = NULL,
  this_line_width = 1,
  this_point_type = NULL,
  this_col = "grey",
  otl_col = NULL,
  this_plot_cex = 0.8,
  this_lab_cex = NULL,
  this_axis_cex = NULL,
  this_mar = c(4, 4, 4, 0.5),
  this_reset = FALSE
)
```

Arguments

seas_obj	A seas object for a single series generated from the seasonal package (Sax and Eddebuettel 2018). This is a required entry.
this_table	Character string; X-13ARIMA-SEATS table name or abbreviation. If not a valid table name, the function will print an error message and return a NULL. This entry is required.
main_title	Character string; main title of plot. Default is 'Cumulative periodogram'.
main_title_line	Integer scalar; position of main title of plot. Default is 2.75.
main_title_cex	Numeric scalar; scaling for main title of plot. Default is this_plot_cex + 0.1.
x_axis_label	Character string; x-axis label for plot, if specified.
y_axis_label	Character string; y-axis label for plot, if specified.
x_axis_main_ticks	Numeric vector; X-axis tick marks. Default is determined by the axis function.
y_axis_main_ticks	Numeric vector; Y-axis tick marks. Default is determined by the axis function.
y_limit	Numeric vector of length 2; Range of values you wish the plot to be plotted over. Default is range of the series specified.
start_plot	Integer vector of length 2; Starting date for plot. Default is starting date for the time series.
do_grid	Logical scalar; indicates if certain plots will have grid lines. Default is no grid lines.
draw_recess	Logical scalar; indicates if certain plots will have shaded areas for NBER recession dates (NBER 2026). Default is no recession shading.
recess_start	numeric matrix; Rows of dates for additional recession starting and ending dates. Default is not to add recession dates.
recess_col	Character string; color used for shading of recession region. Default is 'lightgrey'.
recess_sub	Logical scalar; indicates if x-axis label for recession is produced for this plot. Default is x-axis label is produced
add_otl	Logical scalar; indicates if lines for identified outliers are included in series plots. Default is not including lines for identified outliers.
use_ratio	Logical scalar; indicates if plots of seasonal factors, irregular, and residuals are done as ratio plots. Default has these plots as time series line plots.
add_sub_title	Logical scalar; indicates if plots will include subtitles denoting what series are plotted. Default is not including subheaders.
sub_title_line	Integer scalar; position of main title of plot. Default is 1.
sub_title_cex	Numeric scalar; scaling for main title of plot. Default is this_plot_cex - 0.1.
this_line_type	Integer scalar; indicates line type of plot. Default is 1.
this_line_width	Integer scalar; line width used for series. Default is 1.
this_point_type	Integer scalar; point type used for series. Default is no points plotted.
this_col	Character scalar; color used for plot. Default is 'grey'.
otl_col	Character array of length 6; color used for different outliers, with the order being 'ao', 'ls', 'tc', 'so', 'rp', 'tls'. Default is colors from the RColorBrewer palette "Paired" (Neuwirth 2022).

this_plot_cex	Numeric scalar; scaling for the plot itself. Default is 0.8.
this_lab_cex	Numeric scalar; scaling for plot labels. Default is the value of this_plot_cex.
this_axis_cex	Numeric scalar; scaling for plot axis. Default is the value of this_plot_cex.
this_mar	Numeric vector; set margins for the plot. Default is c(4, 4, 4, 0, 5).
this_reset	Logical scalar; if TRUE, the values of par are reset. Default is FALSE.

Details

Version 1.5, 2026-04-08

Value

Generate plot of user-specified series. If series not specified, print out error message and return NULL.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

NBER (2026). “Business Cycle Dating.” <https://www.nber.org/research/business-cycle-dating>.

Neuwirth E (2022). *RColorBrewer: ColorBrewer Palettes*. R package version 1.1-3, <https://CRAN.R-project.org/package=RColorBrewer>.

Sax C, Eddelbuettel D (2018). “Seasonal Adjustment by X-13ARIMA-SEATS in R.” *Journal of Statistical Software*, **87**(11), 1–17. doi:10.18637/jss.v087.i11.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = "(0 1 1)(0 1 1)",
    x11="", x11.save = c("e18"))
blstinplot::plot_table(air_seas, "d11",
  y_axis_label = "AirPassengers",
  main_title = "X-11 Seasonal Adjustment of Airline Passengers",
  main_title_cex = 1.0,
  add_sub_title = TRUE,
  do_grid = TRUE,
  draw_recess = TRUE,
  use_ratio = FALSE,
  add_otl = TRUE,
  this_col = "black")
blstinplot::plot_table(air_seas, "e18",
  y_axis_label = "AirPassengers",
  main_title = "Adjustment Factors of Airline Passengers",
  main_title_cex = 1.0,
  add_sub_title = TRUE,
  do_grid = TRUE,
  draw_recess = TRUE,
  use_ratio = TRUE,
  add_otl = TRUE,
  this_col = "darkblue")
```

proc_outlier	<i>Extract dates from outlier text</i>
--------------	--

Description

Process name of outlier regressor to extract the dates associated with the outlier.

Usage

```
proc_outlier(this_outlier = NULL, this_freq = 12, add_type = TRUE)
```

Arguments

this_outlier	Character string; outlier regressor. This is a required entry.
this_freq	integer scalar; time series frequency. Default is 12.
add_type	logical scalar; determines if type of outlier is added to the output. Default is TRUE.

Details

Version 2.2, 2026-02-26

Value

List of either year and month/quarter of outlier, or year and month/quarter of start and end of outlier

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    transform.function = "log",
    arima.model = "(0 1 1)(0 1 1)",
    regression.aictest = 'td',
    forecast.maxlead=36,
    check.print = c( "pacf", "pacfplot" ),
    x11="",
    slidingspans = "",)
this_auto_outlier <- get_auto_outlier_string(air_seas)
this_outlier      <- proc_outlier(this_auto_outlier)
```

reset_par	<i>Reset</i> par()
-----------	--------------------

Description

Reset graphics parameters for plots; taken from Stack Overflow (2012).

Usage

```
reset_par(old_par = NULL)
```

Arguments

old_par List object; old graphical parameters saved from par. This is a required entry.

Details

Version 1.1, 2026-04-07

Value

returns reset graphics parameters

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

References

Stack Overflow (2012). “Reset the graphical parameters back to default values without use of dev.off()” <https://stackoverflow.com/questions/9292563/reset-the-graphical-parameters-back-to-default-values-without-use-of-dev-off> [Accessed 01-04-2026].

Examples

```
old.par <- par(no.readonly = TRUE)
par(mar=c(5.1, 3.1, 4.1, 1.1), mfrow=c(2,2))
xt_names <- names(xt_data_list)
for (i in 1:4) {
  plot(xt_data_list[[i]], main = xt_names[i], type="l")
}
reset_par(old.par)
```

set_day_and_month	<i>generate date object from days and year</i>
-------------------	--

Description

Generate a date object from the number of days from the beginning of the year and the target year.

Usage

```
set_day_and_month(day_number, target_year)
```

Arguments

day_number	Integer scalar, number of days from the start of the year. This is a required entry.
target_year	Integer scalar, year of the date object. This is a required entry.

Details

Version 1.1, 1/7/2026

Value

Date object determined by the input arguments.

Author(s)

Brian C. Monsell, <monsell.brian@bls.gov> or <bcmonsell@gmail.com>

Examples

```
this_start <- set_day_and_month(1, 2001)
this_end   <- set_day_and_month(350, 2025)
```

shoes2008	<i>Retail sales of shoes, 2008</i>
-----------	------------------------------------

Description

A time series object containing retail sales of shoes from the US Census Bureau.

Usage

```
data(shoes2008)
```

Format

Retail sales of shoes ending in April of 2008.

visual_sig_peaks	<i>Flag visual significant peaks in spectra</i>
------------------	---

Description

Determine positions of visual significant peaks in spectra

Usage

```
visual_sig_peaks(seas_obj = NULL, spec_type = "sa", spec_freq_code = "seas")
```

Arguments

seas_obj	seas object generated from a call of seas on a single time series. This entry is required.
spec_type	Character string; type of spectrum. Possible values are 'ori', 'irr', 'rsd', 'sa', 'comp', 'indsa', 'indirr', 'extrsd'. Default is 'sa'.
spec_freq_code	Character string; type of frequency being tested. Possible values are 'seas' or 'td'. Default is 'seas'.

Details

Version 1.0, 2026-04-08

Value

If visually significant peaks found, a numeric vector of the position of the peak frequencies. If no peaks found, 0 is returned.

Examples

```
air_seas <-
  seasonal::seas(AirPassengers,
    arima.model = '(0 1 1)(0 1 1)',
    x11='')
vp_ori_seas <-
  visual_sig_peaks(air_seas, spec_type = 'ori')
vp_ori_td <-
  visual_sig_peaks(air_seas,
    spec_type = 'sa',
    spec_freq_code = 'td')
```

xt_data_list*US Building Permits, One Family Units*

Description

A list object of US One family Building Permits from the US Census Bureau for four regions expressed as time series objects that end in October, 2006.

Usage

```
data(xt_data_list)
```

Format

A list object with 4 time series elements:

mw1u Midwest one family building permits

ne1u Northeast one family building permits

so1u South one family building permits

we1u West one family building permits

Index

- * **datasets**
 - employment_data_mts, [8](#)
 - employment_list, [8](#)
 - shoes2008, [27](#)
 - xt_data_list, [29](#)
- add_outlier_lines, [2](#)
- cnv_color_codes, [3](#)
- color_blind_palette, [4](#)
- convert_date_to_tis, [5](#)
- display_color, [6](#)
- draw_recession, [7](#)
- employment_data_mts, [8](#)
- employment_list, [8](#)
- flag_peak, [9](#)
- gen_tick_positions, [10](#)
- get_auto_outlier_string, [11](#)
- get_month_index, [12](#)
- get_recession_dates, [12](#)
- get_reg_string, [13](#)
- get_udg_index, [14](#)
- plot_ratio, [15](#)
- plot_series, [17](#)
- plot_sf, [20](#)
- plot_table, [22](#)
- proc_outlier, [25](#)
- reset_par, [26](#)
- set_day_and_month, [27](#)
- shoes2008, [27](#)
- visual_sig_peaks, [28](#)
- xt_data_list, [29](#)